RÉPUBLIQUE ALGÉRIENNE DÉMOCRATIQUE ET POPULAIRE

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR ET DE LA RECHERCHE SCIENTIFIQUE



ÉCOLE NATIONALE SUPÉRIEURE D'INFORMATIQUE

EX. INI (INSTITUT NATIONAL DE FORMATION EN INFORMATIQUE)

AI Comparison Evidence

Civic Pulse

Turning NYC 311 requests into decisions for city managers

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Advanced Database Systems · SID2 · 2026/2027

1. Name of the AI tool used

Model name	Claude (Anthropic), used in a new chat with no prior context
Date	08/11/2026

2. Exact prompt submitted

What we gave the tool:

- The raw data file: raw_311_2024_sample.csv (45,750 requests from 2024).
- The same problem statement and the same four questions as in our own project (prompt.txt).
- No hints about data problems: we did not mention missing dates, the empty due_date or the DHS closing dates.

Exact text of the prompt:

You are a data analyst. I give you a CSV file with 45,750 NYC 311 service requests from 2024 (a 2% sample).

Problem: New York city managers must decide what to fix first, where to send their teams, and when they need more staff.

Build a dashboard with 3 pages that answers these 4 questions:

1. What are the most common problems, and which ones are growing?
2. Where in the city are the most requests and the most unsolved ones?
3. When do requests arrive (months, days, hours)?
4. How fast does each agency solve requests, and which is slowest?

Page 1: Overview. Page 2: Geography. Page 3: Performance.

Then give 3 findings, each with a recommendation for city managers.

The four questions the dashboard had to answer (from our project plan; check that they match prompt.txt):

- What are the most common problems, and which ones are growing?
- Where in the city are the most requests and the most unsolved ones?
- When do requests arrive (months, days, hours)?
- How fast does each agency solve them, and who is slowest?

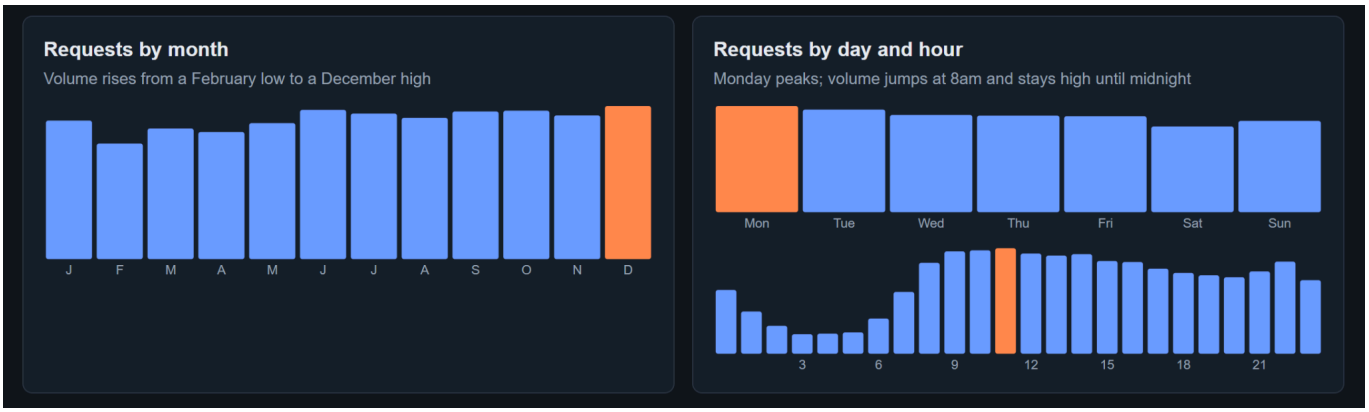
3. Generated dashboard

- Claude produced a 3-tab HTML dashboard (Overview, Geography, Performance) and three findings with recommendations. Published at: https://claude.ai/artifact/5VVSXkZcjkbntSt1UFVrK6

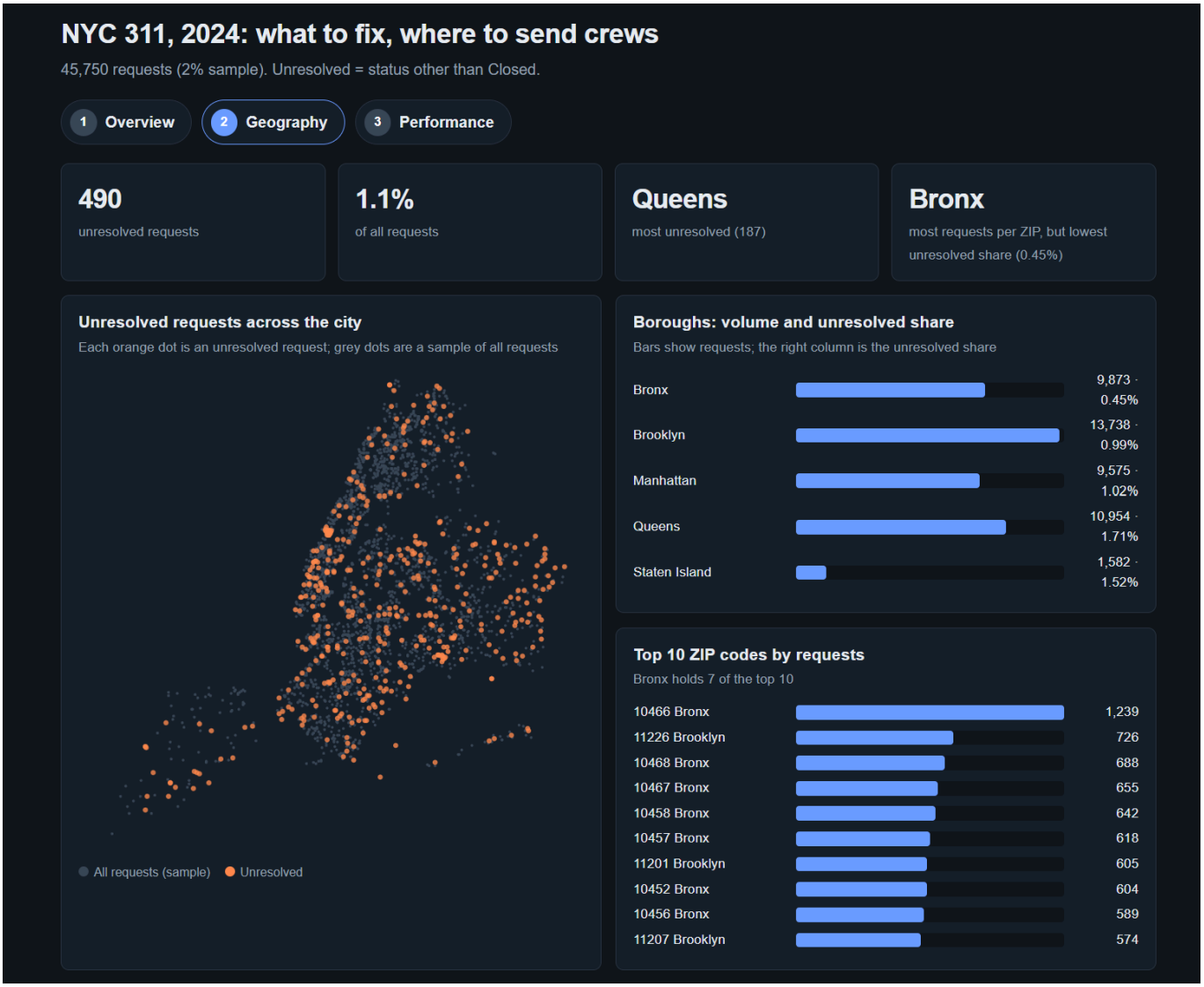
- Claude said it did not render or check the page after building it. It counted a request as unresolved when its status is not Closed (490 requests, 1.1%) and left out the 10 requests closed before they were created.



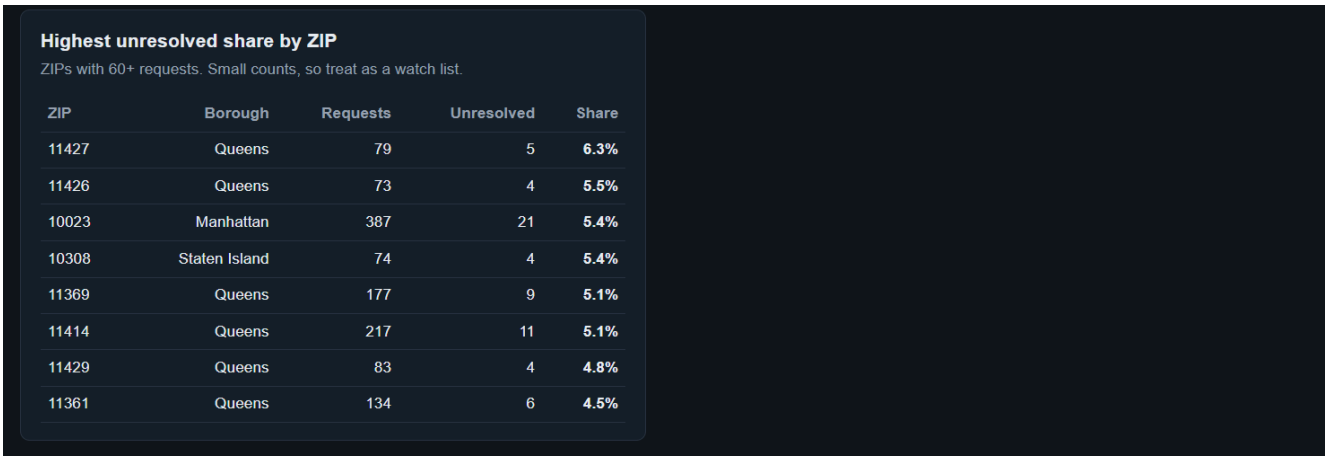
Overview tab: key numbers, most common and fastest growing problems



Overview tab: requests by month, day and hour



Geography tab: map of unresolved requests, boroughs, top ZIP codes

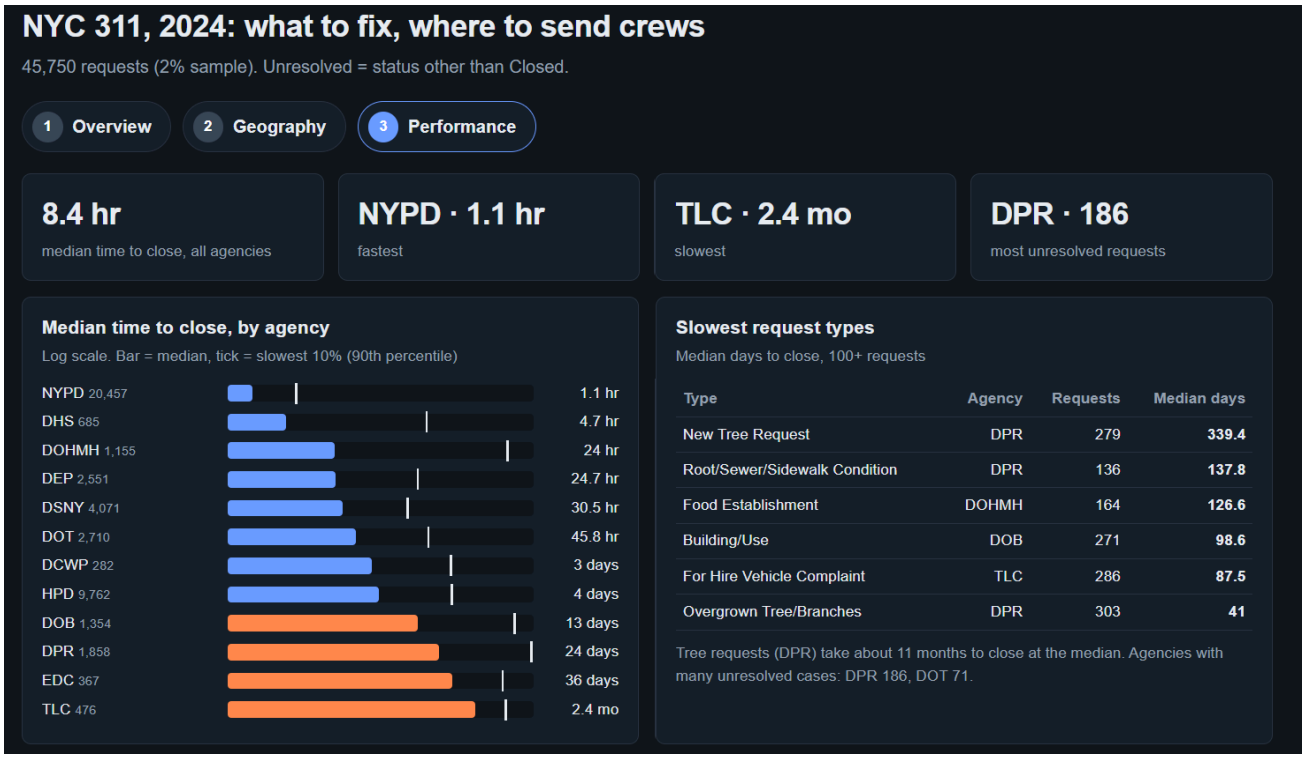


Highest unresolved share by ZIP

ZIPs with 60+ requests. Small counts, so treat as a watch list.

ZIP	Borough	Requests	Unresolved	Share
11427	Queens	79	5	**6.3%**
11426	Queens	73	4	**5.5%**
10023	Manhattan	387	21	**5.4%**
10308	Staten Island	74	4	**5.4%**
11369	Queens	177	9	**5.1%**
11414	Queens	217	11	**5.1%**
11429	Queens	83	4	**4.8%**
11361	Queens	134	6	**4.5%**

Geography tab: ZIP codes with the highest unresolved share

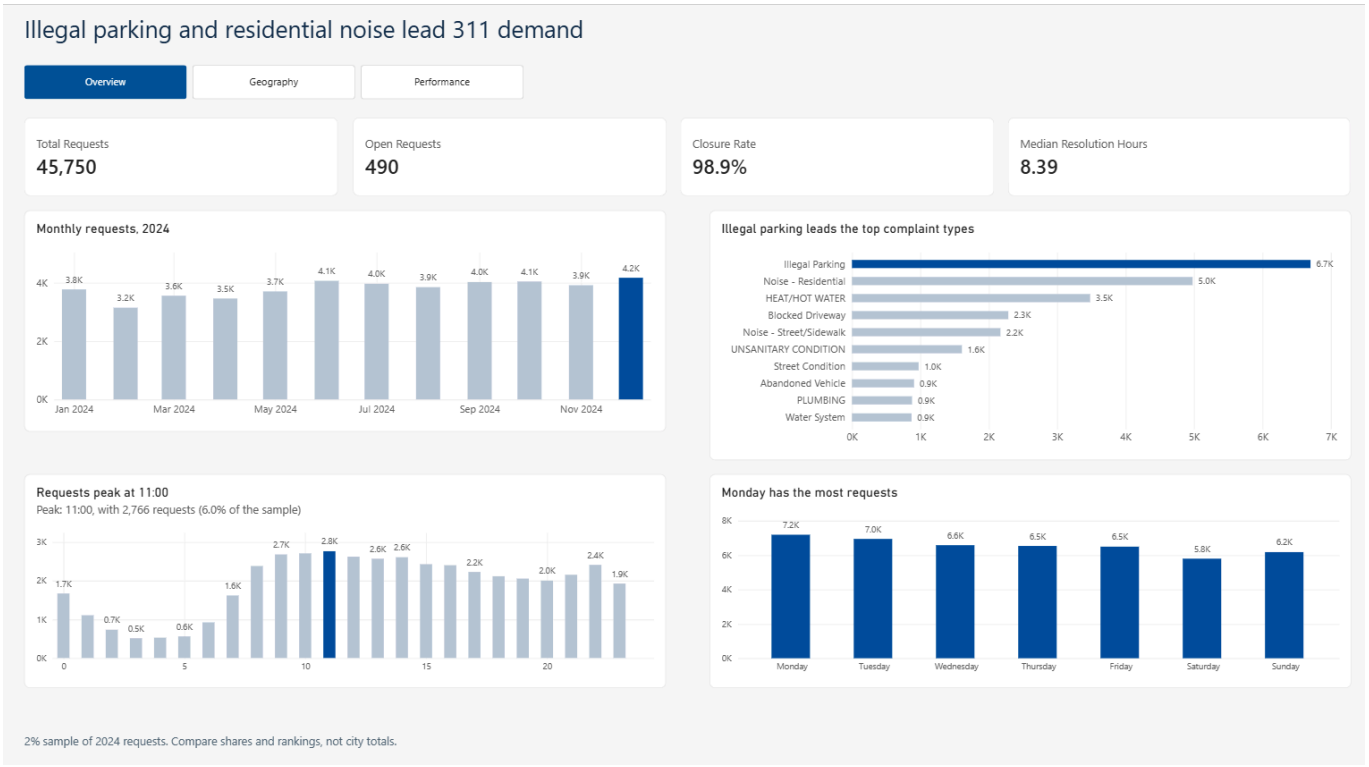


Performance tab: median time to close by agency, slowest request types

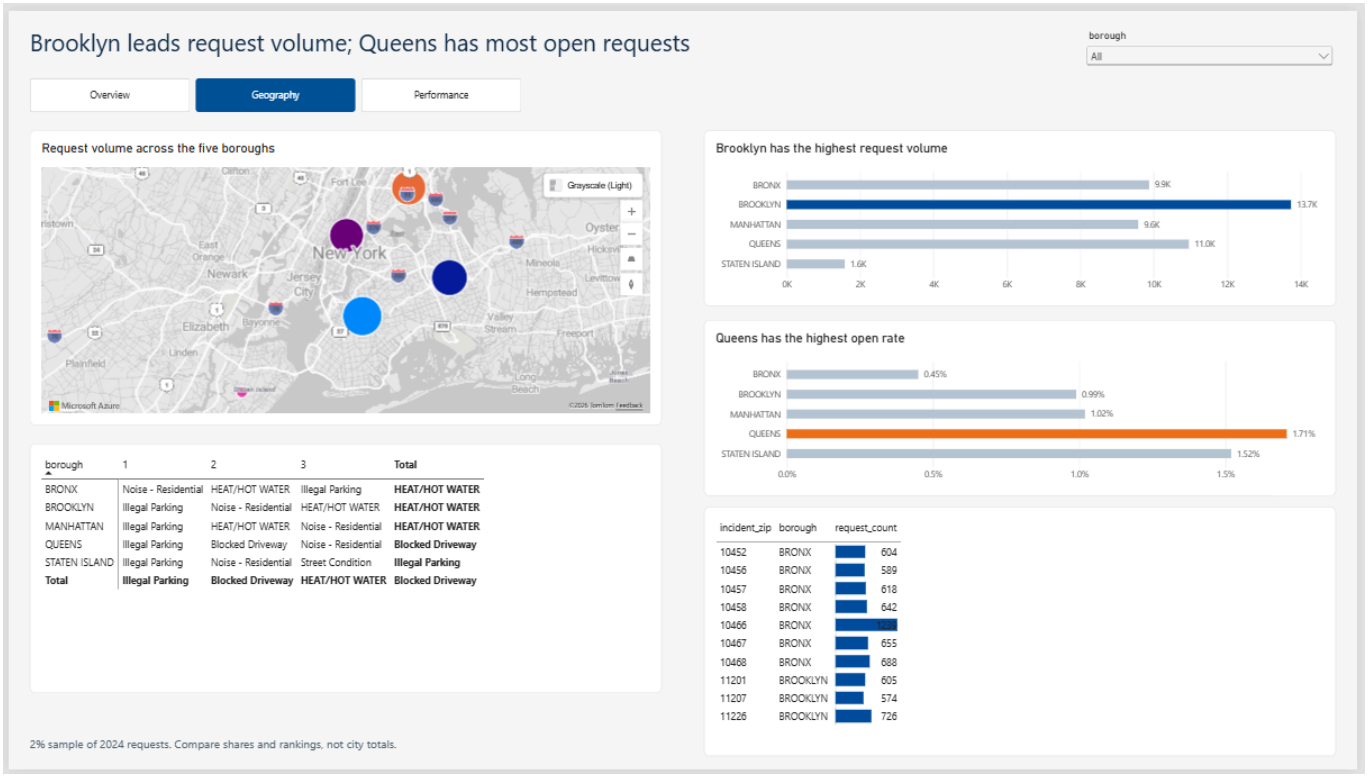
The three findings Claude proposed

- Noise and lead complaints are growing; 311 volume is highest on weekdays from 8am to midnight. Recommendation: move noise and parking shifts to Monday and 8am–midnight, and ask HPD for a lead-inspection plan.
- Queens has the most unresolved requests, the Bronx has the most requests. Recommendation: send crews to the Bronx in proportion to volume, and add follow-up capacity in Queens.
- DPR and TLC are the slowest agencies; parks tree requests are the slowest type. Recommendation: set a service target and a published backlog count for DPR tree requests, and review TLC and EDC.

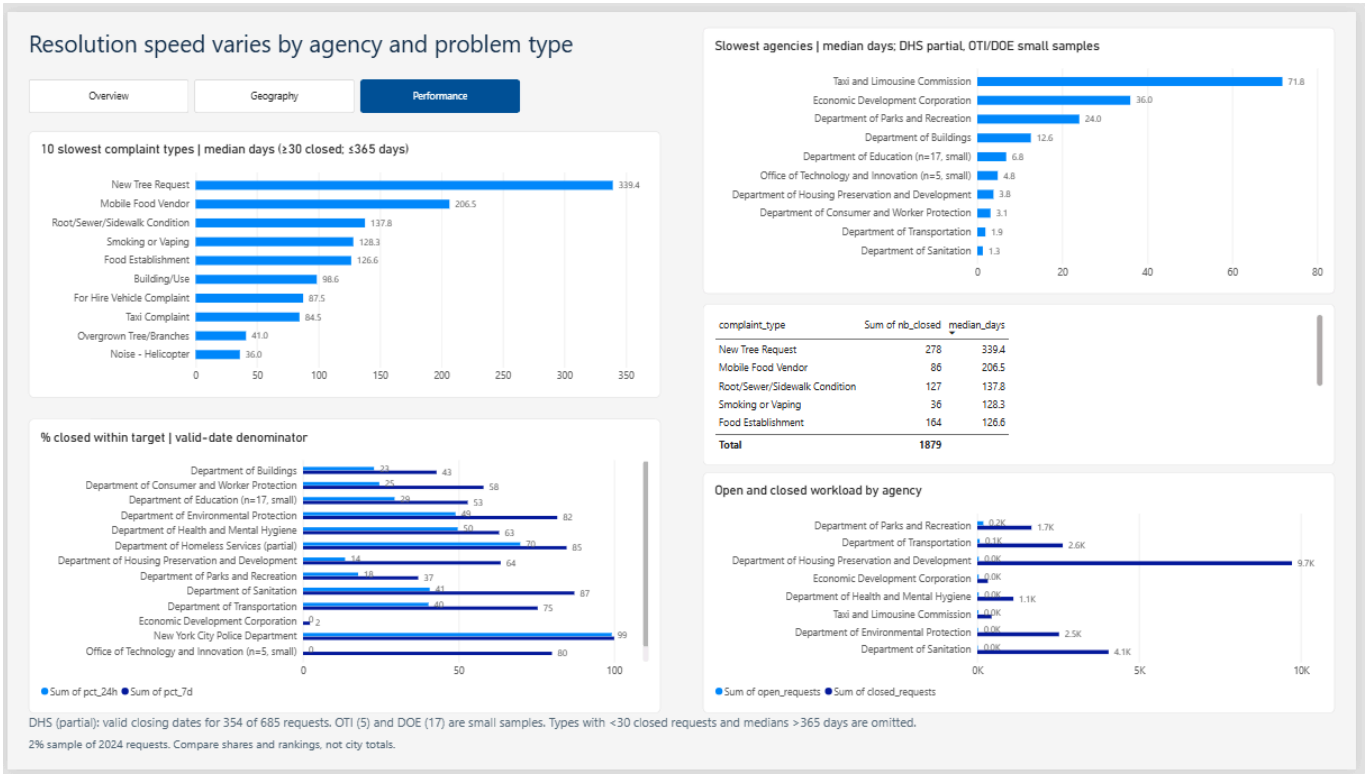
4. Critical comparison with our own dashboard



Our Power BI dashboard: Overview



Our Power BI dashboard: Geography



Our Power BI dashboard: Performance

1. Are its numbers right?

Yes, for almost everything. We checked the AI's figures against our SQL results:

AI claim	Our SQL	Result
Overall median 8.4 h	8.4 h	Match
Bronx 9,873 (0.45% unresolved)	9,873 (0.45%)	Match
Brooklyn 13,738 (0.99%)	13,738 (0.99%)	Match
Manhattan 9,575 (1.02%)	9,575 (1.02%)	Match
Queens 10,954 (1.71%), 187 unresolved	10,954 (1.71%), 187	Match
Staten Island 1,582 (1.52%)	1,582 (1.52%)	Match
Lead 85 to 268 (+215%)	85 and 268	Match
February 3,150, December 4,179	3,150 and 4,179	Match
"The Bronx has the most requests"	Brooklyn is first (13,738 vs 9,873)	Wrong: the Bronx only has the most requests per ZIP code

The AI's numbers were accurate. Its error was in reading its own chart. Its advice to send crews to the Bronx rests on that error.

2. Did it notice the data problems?

- **Yes:** it excluded the 10 requests closed before they were created, defined "unresolved", mentioned the 2% sample and flagged small ZIP counts as a watch list.
- **No:** it missed the 331 DHS requests marked Closed without a closing date, so it presents DHS "about 5 hours" as reliable. It also missed the empty due_date and the requests with no borough or ZIP.

3. Are its recommendations useful?

- **Good:** the DPR tree request target and published backlog count is specific, and it adds a caveat about the sample size.
- **Weak:**
    - "The lead backlog will keep growing" is not supported, because HPD has only 0.4% of its requests open.
    - More noise and parking capacity ignores that the NYPD already closes 99% within 24 hours, so the problem is volume, not speed.
    - "Highest on weekdays" is overstated, because weekends are only slightly lower.

4. Comparison table

Criterion	Our dashboard	AI tool (Claude)	Comment
Numbers correct	All match the SQL	Match on all checked values; 1 wrong claim (Bronx)	Accurate calculations, weaker at reading its own charts
Data problems	DHS dates, bad dates, empty due_date, missing ZIP and borough	Only the bad dates and the sample size	Human review found the DHS problem
Recommendations	Based on all 4 questions, with caveats	Specific, but some unsupported	Needs human checking
Extras	% solved within 24 h and 7 days	Dot map, "slowest 10%" marker, growth by complaint type	Our prompt did not ask for % within 24 h

5. Where our own dashboard is weaker

- Our Overview page shows monthly volume but not which complaint types are growing. The AI answered this with its January–June vs July–December comparison.
- Our Geography page has no map of unresolved requests like the AI's dot map.

5. Conclusion

The AI-generated dashboard was fast and mostly accurate. Claude produced a polished three-page dashboard in minutes, and every figure we checked matched our SQL results. It also covered things our dashboard does not: growth by complaint type and a map of unresolved requests.

It could not replace the human work, though, for three reasons:

- **Missed data problem:** It missed the most important one, the 331 DHS requests marked Closed without a closing date. It therefore presented DHS's median of about 5 hours as reliable, when it covers only 354 of 685 requests.
- **Misread chart:** It misread its own chart, saying the Bronx has the most requests when Brooklyn does (13,738 vs 9,873). Its advice to send crews to the Bronx rests on that error.
- **Unsupported recommendations:** Some were not backed by the data. For example, it predicted a growing lead backlog although HPD has only 0.4% of its requests open.

Our dashboard is slower to build, but it is more reliable. It uses shared SQL rules, documents the data limits (DHS dates, empty due_date, missing ZIP codes and boroughs) and keeps every number traceable to a query.

AI works best as a fast first draft or a second opinion. Before any decision is made, a person must check the data quality, verify the numbers and test whether each recommendation is really supported.